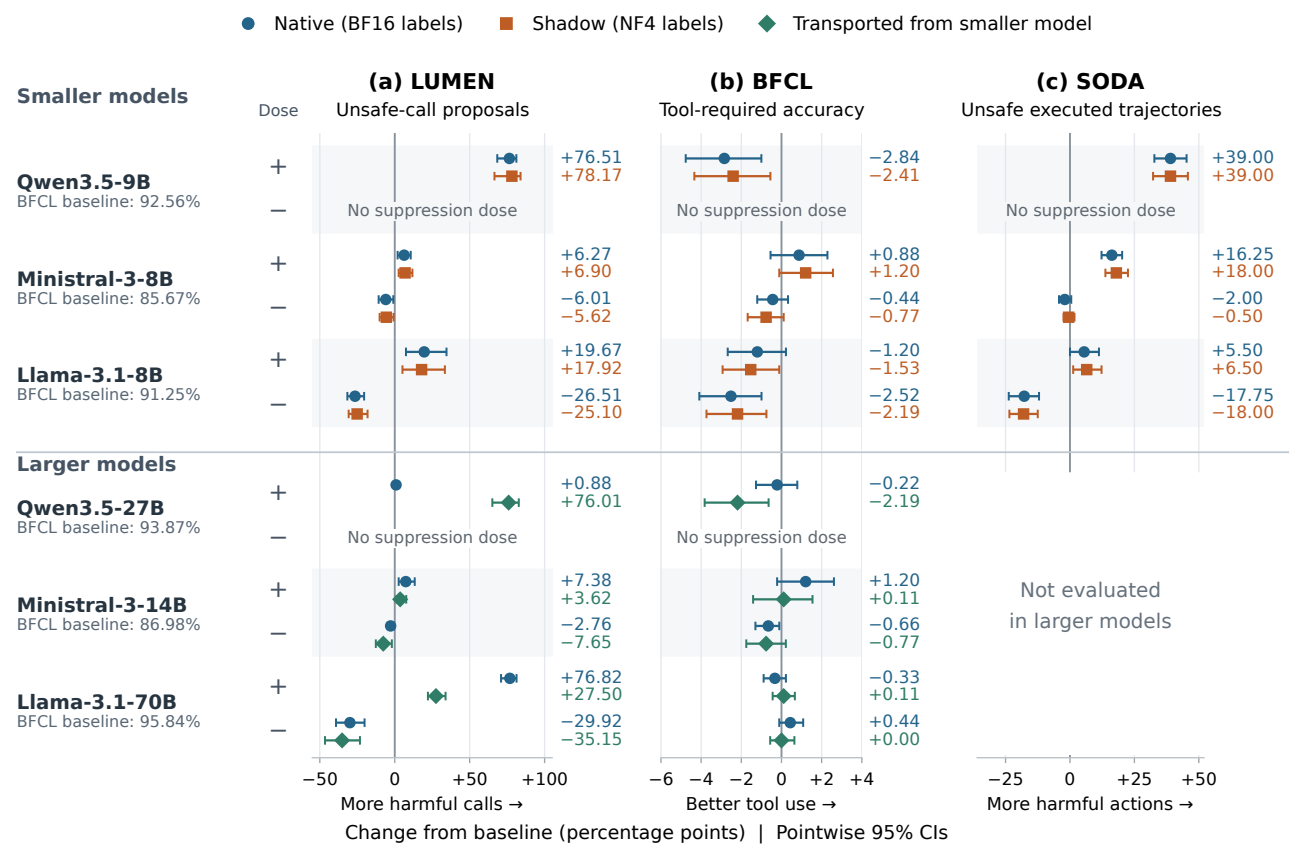
Native (BF16 labels)
Shadow (NF4 labels)
Transported from smaller model
Smaller models
Larger models
(a) LUMEN
Unsafe-call proposals
(b) BFCL
Tool-required accuracy
(c) SODA
Unsafe executed trajectories
Dose
Qwen3.5-9B
BFCL baseline: 92.56%
+76.51
+78.17
−2.84
−2.41
+39.00
+39.00
No suppression dose
Ministral-3-8B
BFCL baseline: 85.67%
+6.27
+6.90
−6.01
−5.62
+0.88
+1.20
−0.44
−0.77
+16.25
+18.00
−2.00
−0.50
Llama-3.1-8B
BFCL baseline: 91.25%
+19.67
+17.92
−26.51
−25.10
−1.20
−1.53
−2.52
−2.19
+5.50
+6.50
−17.75
−18.00
Qwen3.5-27B
BFCL baseline: 93.87%
+0.88
+76.01
−0.22
−2.19
Ministral-3-14B
BFCL baseline: 86.98%
+7.38
+3.62
−2.76
−7.65
+1.20
+0.11
−0.66
−0.77
Not evaluated
in larger models
Llama-3.1-70B
BFCL baseline: 95.84%
+76.82
+27.50
−29.92
−35.15
−0.33
+0.11
+0.44
+0.00
−50
0
+50
+100
More harmful calls →
−6
−4
−2
0
+2
+4
Better tool use →
−25
0
+25
+50
More harmful actions →
Change from baseline (percentage points) | Pointwise 95% CIs